THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The ability of the eye to see objects clearly at different distances depends on 1

(a) the pupil moving the lens
(b) the retina moving back and forth
(c) the ciliary muscles changing the shape of the lens
(d) the cornea changing its colour

2. A cylindrical wire of resistance R is cut into n equal parts which are joined in parallel. The equivalent resistance is 1

(a) n^2 R **(b)** R / n **(c)** R / n^2 **(d)** n R

3. A bar magnet is brought towards a closed coil of wire. The coil 1

(a) develops an induced current while the magnet moves
(b) develops a current even if the magnet is at rest
(c) becomes a permanent magnet
(d) never develops a current

4. In a food chain: plants, insects, frogs, snakes, eagles, the primary and the quaternary consumers are 1

(a) insects and eagles **(b)** snakes and eagles
(c) plants and insects **(d)** frogs and snakes

5. On adding dilute HCl to a solid X, a colourless gas evolves that turns lime water milky. X is likely: 1

(a) Na_2CO_3 **(b)** Na_2SO_4 **(c)** $NaCl$ **(d)** $NaNO_3$

6. A compound on heating gives CO_2; its aqueous solution is mildly basic and is used in baking. It is: 1

(a) Na_2CO_3 **(b)** NaOH **(c)** $NaHCO_3$ **(d)** NaCl

7. A black powder X when heated with carbon gives brown metal Y which on heating in air returns to X. Y is: 1

(a) iron **(b)** copper **(c)** mercury **(d)** sodium

8. Compound Z on heating with conc. H_2SO_4 gives ethene and on oxidation gives ethanoic acid. Z is: 1

(a) ethanol **(b)** ethanal **(c)** ethane **(d)** methanol

9. Two potted plants are kept in closed chambers: one in the dark, the other in light with a dish of KOH solution (which absorbs carbon dioxide). Starch will be absent in: 1

(a) both plants **(b)** only the plant in the dark
(c) only the plant with KOH **(d)** neither plant

10. Overproduction of growth hormone in childhood causes: 1

(a) dwarfism **(b)** diabetes **(c)** gigantism **(d)** goitre

11. A woman's oviducts are blocked. Which would be affected most? 1

(a) menstruation of the male **(b)** The egg cannot meet the sperm
(c) production of eggs **(d)** production of sperm

12. A farmer saves seeds from tall plants raised in good soil. The next generation is grown in poor soil and is short. This shows 1

(a) blending
(b) environment affects expression of the trait
(c) genes were lost
(d) tall is recessive

13. A lens produces an erect image 4 times the size of an object placed 7.5 cm from it. Then f is 1

(a) -10 cm **(b)** +5 cm **(c)** +7.5 cm **(d)** +10 cm

14. The size of the particles in the atmosphere responsible for the white colour of clouds is 1

(a) zero **(b)** larger than the wavelength of light
(c) about the size of an atom **(d)** much smaller than the wavelength

15. For the given resistors of 3 ohm, 6 ohm connected in parallel and then in series with 2 ohm and a 6 V cell, the current is 1

(a) 1 A **(b)** 0.75 A **(c)** 1.5 A **(d)** 3 A

16. The average value of a pure alternating current over one complete cycle is 1

(a) zero **(b)** half of maximum **(c)** equal to peak **(d)** maximum

17. Which of the following does NOT contribute to biomagnification? 1

(a) Non-degradable pesticides **(b)** DDT

(c) Biodegradable waste **(d)** Mercury

18. Assertion (A): A balanced chemical equation obeys the law of conservation of mass. Reason (R): The number of atoms of each element is equal on both sides. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. Assertion (A): Sodium hydrogen carbonate is used as an antacid. Reason (R): It reacts with HCl in the stomach to release CO_2 and neutralise excess acid. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. Assertion (A): Metals are good conductors of electricity. Reason (R): Metals have free electrons. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

21. Assertion (A): Soap does not form lather in hard water. Reason (R): Hard water contains calcium and magnesium ions which form insoluble salts with soap. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

22. Assertion (A): Xylem transports water upward in tall trees. Reason (R): Transpiration creates a suction force that pulls water. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

23. Assertion (A): Iodised salt is recommended for good health. Reason (R): Iodine is required for the synthesis of thyroxine. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

24. Assertion (A): Cross-pollination leads to greater variation. Reason (R): It combines genes from two different plants. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

25. Assertion (A): In a monohybrid cross the F2 genotypic ratio is 1 : 2 : 1. Reason (R): F1 plants form two kinds of gametes in equal numbers. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

26. Assertion (A): A ray of light passing through the optical centre of a lens does not bend. Reason (R): Thickness of the lens at the centre is negligible and faces are parallel. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

27. Assertion (A): The sky appears blue. Reason (R): Blue light is scattered the most by air molecules. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

28. Assertion (A): Parallel combination of resistors is used in houses. Reason (R): Each appliance gets the same voltage and works independently. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

29. Assertion (A): A current-carrying conductor in a field always experiences a force. Reason (R): The magnetic field acts on the moving charges in the conductor. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

30. Assertion (A): Top carnivores contain the highest concentration of pesticides. Reason (R): Pesticides are non-degradable and accumulate along the food chain. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

31. In the electrolysis of water, 20 mL of oxygen is collected. The volume of hydrogen collected is: 1

(a) 60 mL **(b)** 10 mL **(c)** 20 mL **(d)** 40 mL

32. A solution has pH 3. Another solution has pH 6. Which statement is correct? 1

(a) The second is more acidic **(b)** The first is more acidic
(c) Both are equally acidic **(d)** Both are basic

33. A metal oxide X dissolves in water to turn red litmus blue. X is most likely: 1

(a) CuO **(b)** Al_2O_3 **(c)** Na_2O **(d)** Fe_2O_3

34. On complete combustion in air, ethanol gives: 1

(a) ethanoic acid only **(b)** carbon monoxide and hydrogen
(c) carbon dioxide and water **(d)** ethene and water

35. A heart pumps 70 mL of blood per beat at 72 beats per minute. Volume pumped in a minute (in mL) is: 1

(a) 5040 **(b)** 7200 **(c)** 504 **(d)** 720

36. An impulse is blocked in the sensory neuron only. The patient will: 1

(a) have no reflex only **(b)** feel but not move
(c) not feel a stimulus **(d)** move more

37. Starting with one yeast cell, after 4 successive buddings of each cell that each give one new cell (doubling), the number is 1

(a) 8 **(b)** 4 **(c)** 32 **(d)** 16

38. A plant of genotype TtRr produces gametes TR, Tr, tR and tr in equal numbers. The probability of a gamete tr is 1

(a) 1 **(b)** 1/2 **(c)** 1/8 **(d)** 1/4

39. Light of speed 1.5×10^8 m/s in a medium has refractive index 1

(a) 2 **(b)** 1.5 **(c)** 0.5 **(d)** 4.5

40. A second identical prism is placed upside down in contact with the first one, in the path of a narrow white beam. The emergent beam is 1

(a) white light **(b)** a wider spectrum
(c) only violet light **(d)** only red light

41. **Assertion (A):** Both parents contribute equally to the genetic material of a human child. 1
Reason (R): The sex of a human child is determined by the sperm that fertilises the egg.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** A ray of light passing through the centre of curvature of a concave mirror retraces its path after reflection. 1
Reason (R): The radius of curvature of a spherical mirror is twice its focal length.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** A rainbow is seen when the Sun is behind the observer and rain is falling in front. 1
Reason (R): A rainbow is formed by dispersion, internal reflection and refraction of sunlight in water droplets.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The resistance of a conductor is directly proportional to its length. 1
Reason (R): The resistivity of a material depends on its temperature.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** A solenoid carrying current behaves like a bar magnet. 1
Reason (R): The magnetic field inside a long solenoid is uniform.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The use of chlorofluorocarbons (CFCs) has damaged the ozone layer. 1
Reason (R): The ozone layer absorbs the harmful ultraviolet radiation from the Sun.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Rusting of iron is an oxidation process. 1
Reason (R): During rusting, iron is reduced by oxygen and moisture.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Tooth decay starts when the pH of the mouth falls below 5.5. 1
Reason (R): Tooth enamel is dissolved by the bases produced by bacteria in the mouth.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Iron articles are galvanised by coating them with zinc. 1
Reason (R): Zinc is less reactive than iron and so cannot be corroded.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

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50. **Assertion (A):** Vegetable oils are hydrogenated using nickel catalyst to obtain saturated fats. 1
Reason (R): Hydrogenation converts the single bonds of unsaturated oils into double bonds.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**

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